



Report

Project: TravelTide

TravelTide is preparing to launch a new customer reward program. To support its success, I analyzed session-level and booking data to identify user segments and define tailored perks that incentivize program participation.

This report summarizes the approach, key insights, and recommendations for reward design. It avoids technical detail and focuses on actionable findings for marketing and product strategy.

Data Basis

The analysis is based on session-level data between **April 2021 and August 2023**. To focus on active users, the dataset was filtered for sessions **after January 4th, 2023**, and users with more than seven sessions. After filtering, 5,782 users remained.

Data Cleaning

During the data preparation phase, I identified several inconsistencies in the dataset that required correction. For example, some trips were recorded with **negative values** for the number of nights (e.g., “-1 nights”), which is logically not possible. To address this, I recalculated the nights using the return time of the trip.

Missing discount values were also identified and replaced with the average discount values across the dataset to maintain consistency.

Outliers such as very long trips or unusually high spending were retained, as they likely represent real traveler types (e.g., luxury or long-distance travelers) and provide valuable insights for segmentation.

Feature Engineering

To better capture user behavior, I engineered new features that represent averages of key travel attributes. Examples include the average price per kilometer per person & average number of seats booked.

By **focusing on averages**, the analysis highlights the dominant behavior of each user. For instance, if a traveler sometimes flies alone but also takes trips with family, they would not automatically be classified as a “family traveler.” Instead, their overall dominant behavior determines their segment.

In addition to continuous averages, I also derived **binary flags** (0/1 variables) to capture distinct characteristics, such as whether a user has children, is older in age, or frequently travels on weekends. To define these binary features, I applied percentile thresholds (identifying the highest and lowest ranges of behavior) and compared them against each user’s average values.

This method allowed me to transform continuous behavioral patterns into clear True/False indicators, which served as the foundation for the segmentation.



Segmentation Approach

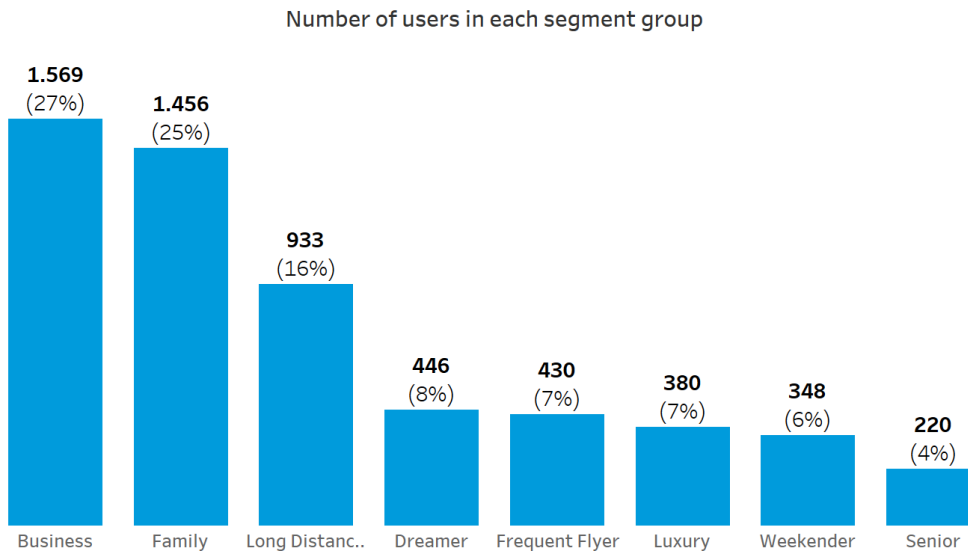
The Segmentation was based on weighted thresholds and percentiles, ensuring clear rules and avoiding overlaps. Each user is assigned to exactly one segment that best represents their dominant behavior.

Why this method?

- Provides **transparent rules** that are easy to interpret and translate into marketing actions.
- Guarantees that each user belongs to **exactly one segment**, without the need for a generic “Others” group.
- Uses **explicit thresholds** (e.g., above the 90th percentile of flights = frequent flyer) to draw clear boundaries, rather than relying on continuous “more or less” values.
- Ensures the segments reflect **real and recognizable traveler patterns**, not just statistical clusters.

Segments identified (8 total):

Segment	Key Features	Reward Potential
Business	- Trip during the week - Travelled alone - Has no bags - Low page clicks - Low session time	- flexibility and cancellation options
Family	- Has Children - Groups of ≥ 3 - Travelled with bags - Is married	- Family/Group Deals
Luxury	- Amount of base fare spending bigger than 90% of average spendings - Price per km per person bigger than 90% of average	- exclusive service
Frequent Flyer	- Number of flights bigger than 95% of all flights	- Special offer for high amount of flights booked
Long Distance Traveler	- Average Flight Distance bigger than 3000 km	- Extra Point for Long Distance Flights
Weekender	- Departure on Friday or Saturday - Return on Sunday	- Bundled Weekend Deals
Senior	- Age over 67 - Has bookings	- Local Senior Support
Dreamer	- No bookings	- Low-Barrier entry offers



Key segmentation insights:

- **Strong presence of Business & Family travelers** are the core segments for TravelTide.
- **Luxury & Frequent Flyers & Long Distance Travellers** are smaller in number, but high in spending potential.
- **Weekenders** show lower spending potential due to their short trips but higher frequency, making them attractive for volume-based perks.
- **Dreamers** represent untapped potential if perks are well designed.
- **Seniors** form the smallest group, but often require tailored support services

Reward Perk Recommendations

Each segment was matched with a perk designed to increase reward program sign-up:

- **Family** → Group discounts (e.g., 30% off for 3+ travelers with children)
- **Business** → Instant rescheduling & flexible cancellations
- **Luxury** → Concierge services (e.g. VIP upgrades, transfers)
- **Frequent Flyers** → Streak rewards (e.g. bonus for 5 flights within 6 months)
- **Long Distance Traveler** → Extra points for long-distance flights
- **Dreamer** → Low-cost trial trip (e.g. 50% discount entry offer)
- **Senior** → Local support services with transportation for free
- **Weekender** → Weekend deal bundles with free excursion.

Conclusion & Next Steps

The analysis provides a **strong foundation** for a tailored reward program, but it should be considered a **starting point**. Perks need to be iteratively tested and refined based on user feedback and measured adoption

Next steps recommended:

1. Launch A/B Testing for all Perks in all groups gain insightful data about the fit to the group
2. Track adoption via dashboards (conversion rate, engagement, segment growth).
3. Expand/Optimise perks iteratively, ensuring each segment receives measurable value
4. Implement feedback mechanism (e.g. survey) to collect insights from first hand